Serge E. Parent, Ph.D.

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Profile Summary

I am a quantitative cell and developmental biologist who completed a Ph.D. studying cell-cell adhesion biophysics. In my thesis I found support for an extracellular matrix-based low adhesion state and showed that this combined with cytoskeletal elasticity and turnover are sufficient for describing adhesion as a dynamical system using a combination of mathematical modeling and experimentation, whereas previous biophysical models of cell-cell adhesion had been equilibrium-based. I currently work at the Allen Institute as a Quantitative Cell Biologist where I use the knowledge and skills I gained during my studies to further our understanding of multicellular systems. My research interests include topics in how cells build and maintain tissues, dynamics in biology, and how we can model and make sense of complex, high-dimensional, or data-rich biological systems.

Skills

Scientific code development

- Workflow development and codebase maintenance using Python (*numpy, scipy, pandas, scikit-image, matplotlib, seaborn, uncertainties, statsmodels, pathlib, dask, multiprocessing*).
- High-performance computing (*SLURM, SSH, bash, using multiprocessing and GPUs*).
- Collaborative code development and version control (*Git, GitHub*).
- Virtual environment management (*uv, pdm, conda*).
- Quantitative data analysis, visualization, statistical testing, linear and non-linear regression fitting.
- Image analysis (*FIJI/ImageJ, scikit-image, bioio, Cellpose, ML-based image analysis*).
- Mathematical and computational modeling of biological processes (*geometric models, systems of ODEs*).
- AI-assisted coding familiarity (*VS Code Copilot: various models*) and general-purpose tools (*Claude Cowork*).

Experimental assay development

- Experimental design (*hypothesis development, alternative approach proposals, choosing control conditions*).
- Microscopy (*laser scanning confocal, spinning disk confocal, lightsheet, multiphoton, epifluorescence, dissection, transmission electron; confocal imaging includes multidimensional/XYZCT fluorescence and subsets thereof*).
- Live and fixed cell and tissue sample preparation (*hiPSC and primary cell culture, tissue isolation and dissociation, IHC, ICC, Xenopus and zebrafish embryo microdissection and mRNA microinjection*).
- Basic molecular biology (*DNA linearization and extraction, mRNA transcription*).
- 3D modeling and 3D printing to create custom experimental tools (*Blender, MakerBot*).

Scientific communication

- Scientific article and report preparation (*Word, LaTeX, Inkscape, Adobe Illustrator, FIJI/ImageJ*).
- Dissemination of scientific findings (*PowerPoints and posters*).

Education

- Ph.D. – Cell & Systems Biology (Dev. Biol. focus); *University of Toronto, CA* — Conferred March 2023
  Thesis: Cell-Cell Adhesion Dynamics & Cortical Tensions in Embryonic Tissue Structure & Mechanics.
- B.Sc. – Cell & Molecular Biology; *University of Toronto, CA* — Conferred June 2014

Research & Professional Experience

Scientist 1, allen institute/cell science — Dec 2023 – present

- Designed and implemented image and data analysis workflows including leveraging high-performance computing, combining classical and ML-based segmentation methods, and measuring biologically-relevant features with established and novel methods contributing to one joint-1st-author manuscript (submitted).
- Carried out analyses and interpretations of results and collaboratively prepared figures and manuscripts contributing to one joint-1st-author manuscript (submitted) and one other published manuscript.
- Reviewed and maintained unsupervised ML-based image analysis workflows and stochastic differential equation-based morphological dynamics and cell state estimation workflows.
- Helped maintain project repos to support good coding practices (*type checking with MyPy, linting with Ruff and Black, using Git pre-commit hooks, engaging in code reviews*), both with and without AI assistance.
- Performed assay development with another scientist imaging live hiPSC-derived endothelial cells on a new microscope system to inform our research direction.

- Prepared and presented scientific findings internally and externally at conferences through posters and PowerPoint presentations to disseminate knowledge and promote *allen institute/cell science.*
- Performed microscopy infrastructure work used by other team members including developing code for extracting metadata in a more interpretable format from *3i* microscopy data and, together with a research assistant, establishing and documenting a QC procedure for a newly acquired microscope.

Postdoctoral Fellow, Scientist (Casual Contract), University of Toronto Jan. 2023 – June 2023

- Designed and executed experiments to address reviewer comments including live confocal and compound microscopy (*XYZT, XYT*) of drug-treated cultured cells, fluorescence confocal imaging of ICC samples (*XYZ*).
- Acquired, manipulated, plotted, and ran statistical tests on time series data to determine if a drug affected cell-cell adhesion dynamics between cultured cells.
- Developed and implemented a method to correct for fluorescence light loss in Z-stacks due to light scattering.
- Revised manuscript text and figures to address reviewer concerns/criticisms and resubmit paper.

Ph.D. Candidate, University of Toronto 2014 – 2022

- Designed and executed quantitative microscopy experiments producing multichannel timelapses (*XYZT, XYT*).
- Developed and implemented microscopy image analysis pipelines for reproducible, high-throughput data analysis to better understand cell-cell adhesion, contributing to a 1st-author peer-reviewed publication.
  - Cell identification (*blob detection*), segmentation (*watershed*), region labeling, and edge detection (*outlining segmentations, sobel filters, Canny edge detection*) of images in timelapses.
  - Feature extraction (*edges, ridges, curvatures, lengths, angles, cell centers, fluorescence intensities*).
  - Quality control processes (*overlays of input images and cell outlines and measurements, interactive plots, checking outputs for NaNs/inadmissible data, masking and filtering data*).
- Performed live microscopy, IHC, and ICC to visualize adhesion-relevant molecules and improve our understanding of cell-cell adhesion, contributing to publications.
- Developed and implemented computational, mathematical, and bias-correcting approaches in Python to model and understand microscopy-derived biological data, resulting in publications.
  - Developed a system of differential equations model of cell-cell adhesion showing that elasticity and turnover of the cytoskeleton can explain empirical adhesion kinetics data that I acquired, whereas the previously established model based solely on cytoskeletal contractility could not.
  - Developed geometric models of interstitial spaces within tissues.
  - Corrected for elongation of imaged tissue structures from oblique sectioning of samples.
- Analyzed, visualized, and evaluated data from experiments and modeling to improve our understanding of cell-cell adhesion, contributing to publications.
  - Non-linear (*logistic*) curve fitting and aligning of time series data, orthogonal distance regressions for fitting arcs to features in images, normalizing data, statistically comparing samples.
  - Various basic and advanced plots (*line plots, heatmaps, 3D plots, "straightening" outlines of cells to plot their fluorescence data as kymographs, etc.*), visualizing Z-stacks in 3D (*Imaris, FIJI/ImageJ, Blender*).
- Communicated results to varied audiences through multiple peer-reviewed publications and poster/oral presentations at departmental, institutional, and international meetings.

Teaching Experience

Teaching Assistant, University of Toronto 2014 – 2020

- 6 semesters leading groups of ∼24 students through scientific seminars, lab experiments, and oral presentations (*incl. courses in Developmental Biology, Cell and Molecular Biology*)
- 2 semesters mentoring a 4th-year project student doing experiments I designed, contributing to a manuscript
- 2 semesters organizing and assessing large groups of ∼50–200 students (*Stem Cell Biology course*)

Awards

- (National Award) NSERC C-GSD Alexander Graham Bell Scholarship Sept. 2018 – Dec. 2020
- (Provincial Award) Ontario Graduate Scholarship Jan. 2018 – Aug. 2018
- (Departmental Award) Yoshio Masui Prize in Development, Molecular or Cell Biology Jan. 2018

Publications

- (*Preprint, under revision*) Angelini E.†, C. L. Leveille†, **S. E. Parent**†, R. J. Zaunbrecher†, T. Barszczewski, J. C. Dixon, F. S. Mohammed, B. Morris, J. S. Yu, J. Arakaki, R. J. Dupar, J. H. Edmonds, E. A. Ehlers, C. R. Gamlin, M. J. Hedayati, C. Hookway, J. McCarley, S. Mogre, A. Phan, B. Roberts, E. E. Sanchez, J. P. Thottam, C. S. Wijesooriya, J. Yao, M. L. Kutys, E. Nazockdast, J. Wang, J. A. Theriot, G. Dalgin, S. M.

Rafelski, M. P. Viana. Dynamics of ML-based morphological features indicate a shear stress-dependent bifurcation of hiPSC-derived endothelial cell states. DOI: 10.64898/2026.07.07.736803.

[†] *These authors contributed equally to this work.*

- Hookway C.[†], A. Borensztejn[†], L. K. Harris[†], T. Barszczewski, S. Carlson, G. Dalgin, S. Mishra, E. M. Adams, J. C. Dixon, R. J. Dupar, J. H. Edmonds, E. A. Ehlers, A. J. Ferrante, M. A. Fuqua, C. R. Gamlin, P. Garrison, J. Gopalan, B. W. Gregor, M. J. Hedayati, V. L. Hurless, K. N. Klein, C. L. Leveille, S. L. Meharry, R. Mercado, H. S. Morris, G. Nadarajan, N. Nivedita, S. A. Oluoch, **S. E. Parent**, A. Phan, B. Roberts, A. Samudre, E. E. Sanchez, M. F. Sluzewski, L. S. Snyder, D. J. Thirstrup, H. F. Thorp, J. P. Thottam, J. R. Torvi, G. Turman, M. P. Viana, L. Wilhelm, C. S. Wijesooriya, J. Yao, J. A. Theriot, R. N. Gunawardane, S. M. Rafelski. 2026. A human induced pluripotent stem cell model for the holistic study of epithelial-to-mesenchymal transitions. *Nature Methods* 23: 1411-1423. DOI: 10.1038/s41592-026-03096-9.

  [†] *These authors contributed equally to this work.*
- **Parent, S. E.**, O. Luu, A. E. E. Bruce, R. Winklbauer. 2024. Two-phase kinetics and cell cortex elastic behavior in *Xenopus* gastrula cell-cell adhesion. *Developmental Cell* 59: 141–155. DOI: 10.1016/j.devcel.2023.11.014.
- Fei, Z., K. Bae, **S. E. Parent**, K. Goodwin, G. Tanentzapf, A. E. E. Bruce. 2019. A cargo model of yolk syncytial nuclear migration during zebrafish epiboly. *Development* 146(1): dev169664. DOI: 10.1242/dev.169664.
- **Parent, S. E.**, D. Barua, R. Winklbauer. 2017. Mechanics of fluid-filled interstitial gaps. I. Modeling gaps in a compact tissue. *Biophys. J.* 113(4): 913–922. DOI: 10.1016/j.bpj.2017.06.062.

  * *This work was recommended in Faculty Opinions as being of importance by Francois Fagotto.*
- Barua, D., **S. E. Parent**, R. Winklbuaer. 2017. Mechanics of fluid-filled interstitial gaps. II. Gap characteristics in *Xenopus* embryonic ectoderm. *Biophys. J.* 113(4): 923–936. DOI: 10.1016/j.bpj.2017.06.063.

  * *This work was recommended in Faculty Opinions as being of importance by Francois Fagotto.*
- Winklbauer, R., **S. E. Parent**. 2017. Forces driving cell sorting in the amphibian embryo. *Mech. Dev.* 144(Pt A): 81–91. DOI: 10.1016/j.mod.2016.09.003.
- Luu, O., E. W. Damm, **S. E. Parent**, D. Barua, T. H. L. Smith, J. W. H. Wen, S. E. Lepage, M. Nagel, H. Ibrahim-Gawel, Y. Huang, A. E. E. Bruce, R. Winklbauer. 2015. PAPC mediates self/non-self-distinction during Snail1-dependent tissue separation. *J. Cell Biol.* 208(6): 839–856. DOI: 10.1083/jcb.201409026.

  * *This paper has been featured in a JCB Biosights video, been chosen as part of a JCB Journal Club package, had a commentary article written about it by David Wilkinson, and been recommended in Faculty Opinions as being of special significance in its field by Patrick Tam and Ira Daar.*

Conferences and Workshops

- Angelini E.[†], C. L. Leveille[†], **S. E. Parent**[†*], R. J. Zaunbrecher[†], T. Barszczewski, J. C. Dixon, F. S. Mohammed, B. Morris, J. S. Yu, J. Arakaki, R. J. Dupar, J. H. Edmonds, E. A. Ehlers, C. R. Gamlin, M. J. Hedayati, C. Hookway, J. McCarley, S. Mogre, A. Phan, B. Roberts, E. E. Sanchez, J. P. Thottam, C. S. Wijesooriya, J. Yao, M. L. Kutys, E. Nazockdast, J. Wang, J. A. Theriot, G. Dalgin, S. M. Rafelski, M. P. Viana. (Jul. 23–26, 2026). Dynamics of ML-derived morphological features reveal shear-stress-dependent reorganization of hiPSC-derived endothelial-like cell states. Presented at: Society for Developmental Biology 85$^{\text{th}}$ Annual Meeting. Las Vegas, NV. (U.S. conference; poster presentation).
- Angelini E.[†], **S. E. Parent**[†*], R. Zaunbrecher[†], Ellen Adams, Tiffany Barszczewski, A. Borensztejn, J. Dixon, R. J. Dupar, J. H. Edmonds, M. J. Hedayati, C. Hookway, C. L. Leveille, J. McCarley, B. Morris, G. Nadarajan, E. E. Sanchez, J. P. Thottam, J. S. Yu, Allen Institute for Cell Science, J. Wang, G. Dalgin, J. A. Theriot, S. M. Rafelski, M. P. Viana. (Dec. 6–10, 2025). A dynamical systems framework applied to live cell imaging of hiPSC-derived endothelial-like cells under flow uncovers a first-order phase transition between morphology-based cell states. Presented at: Cell Bio 2025. Philadelphia, PA. (International conference; oral presentation).
- **Parent, S. E.***, A. E. E. Bruce, R. Winklbauer. (Jul. 26–30, 2019). A biophysical analysis of cell-cell adhesion in the *Xenopus* gastrula. Presented at: Society for Developmental Biology 78$^{\text{th}}$ Annual Meeting. Boston, MA. (International conference; poster presentation).
- **Parent, S. E.***, A. E. E. Bruce, R. Winklbauer. (Jul. 13–17, 2017). Exploring tissue separation at Brachet's Cleft at the level of individual cell pairs. Presented at: Society for Developmental Biology 76$^{\text{th}}$ Annual Meeting. Minneapolis, MN. (U.S. conference; poster presentation).
- **Parent, S. E.***, A. E. E. Bruce, R. Winklbauer. (Aug. 4–8, 2016). Exploring the mechanobiology of Brachet's Cleft at the level of individual cell pairs. Presented at: Society for Developmental Biology 75$^{\text{th}}$ Annual Meeting.

Boston, MA. (International conference; poster presentation).
- Participation in Connaught 3D Bioprinting Workshop (Aug. 2016).
- Several other oral presentations of results at departmental events, institutional events, and meetings as needed.

* *Presenting author.*

† *These authors contributed equally to the work.*

Service

- Helped organize StemDev-ECS, a meeting for early career scientists jointly held by *Society for Developmental Biology*, *International Society for Stem Cell Research*, and *allen institute/cell science* (Sept 23–25, 2026).
- Served as a reviewer for the *Allen Institute Next Gen Leaders* program (2025).
- Reviewed 2 papers with Prof. Winklbauer for *Biophysical Journal* and *Mechanisms of Development*.